AI Tool Prompt Engineering Report

Prepared by:

Chriezl Anne G. Guerrero

Bachelor of Science in Civil Engineering / 3B

21st century IT Skills / 16-608

Submitted to:

Gabriel Quibod

March 18, 2026

NAME OF AI TOOL: <u>CHATGPT</u>

PURPOSE/DOMAIN:

ChatGPT is an AI-Tool used to act as an intelligent assistant that helps people understand, create, and solve problems using language**.**

WHY I CHOSE IT:

I chose ChatGPT because it helps users generate, refine, and understand design ideas quickly, making it an effective tool for conceptual planning, learning, and decision-making. It is also useful for civil engineering students who need simple layout concepts references without using complex software like AutoCAD.

ORIGINAL PROMPT

Initial Prompt:

Create a floor plan for a bungalow house.

Output:

🍳 **3. Kitchen**

- At the back or side
- Size: ~2.5m × 3m
- With door to service area/laundry

🛏️ **4. Bedrooms**

Option A: 2 Bedrooms

- Bedroom 1 (Master): ~3m × 3.5m
- Bedroom 2: ~3m × 3m

Option B: 3 Bedrooms (compact)

- Slightly smaller rooms (~2.5m × 3m each)

🚿 **5. Toilet & Bath**

- Shared CR
- Size: ~1.5m × 2m

🚪 **6. Service/Laundry Area**

- At the back of kitchen
- Optional but very useful

🔹 **Total Floor Area:**

- Around **60–80 sqm** (common for bungalow homes)

🧱 **Room Arrangement**

🛋️ **1. Living Room**

- Located near the **main entrance**
- Size: ~3m × 4m
- Purpose: Receiving guests, relaxation

🍽️ **2. Dining Area**

- Beside the living room (open layout)
- Size: ~3m × 3m
- Connects directly to kitchen

AI-ASSISTED PROMPT (Chatgpt)

Design a functional and space-efficient floor plan for a 60–80 sqm single-story bungalow house in a tropical setting (e.g., Philippines). The layout should include 2–3 bedrooms, 1 toilet and bath, a living area, dining area, kitchen, and a service/laundry area.

Output:

General Information

- **Type:** Single-Storey Bungalow House
- **Total Floor Area: 70 sqm (approx.)**
- **Lot Assumption:** 10m × 12m
- **Occupancy:** Small family (3–5 persons)
- **Design Concept:** Space-efficient, tropical residential layout

Zoning Concept

Zone	Spaces Included	Purpose
Public Zone	Living Room, Dining Area	Social interaction
Private Zone	Bedrooms	Rest and privacy
Service Zone	Kitchen, T&B, Laundry	Utility functions

1. Living Room (Public Zone)

- **Dimensions:** 3.5m × 4.0m (14 sqm)
- **Location:** Front (main entrance)
- **Features:**
  - Large window for natural light
  - Direct connection to dining
  - Space for sofa set + TV console

2. Dining Area (Public Zone)

- **Dimensions:** 3.0m × 3.0m (9 sqm)
- **Features:**
  - Centrally located
  - 4–6 seater dining table
  - Accessible to kitchen and living

3. Kitchen (Service Zone)

- **Dimensions:** 2.5m × 3.0m (7.5 sqm)
- **Layout:** L-shaped counter
- **Features:**
  - Sink, stove, refrigerator space
  - Overhead cabinets
  - Rear door to service area

4. Toilet & Bath (Service Zone)

- **Dimensions:** 1.5m × 2.0m (3 sqm)
- **Fixtures:**
  - Water closet
  - Lavatory
  - Shower area
- **Location:** Central (shared access)

5. Bedrooms (Private Zone)

Bedroom 1 (Master Bedroom)

- **Dimensions:** 3.5m × 3.5m (12.25 sqm)
- **Features:**
  - Queen-size bed
  - Built-in wardrobe
  - Window for cross ventilation

Bedroom 2

- **Dimensions:** 3.0m × 3.0m (9 sqm)
- **Features:**
  - Double/single bed
  - Study table

Bedroom 3 (Optional)

- **Dimensions:** 2.5m × 3.0m (7.5 sqm)
- **Use:** Guest room / study / office

6. Service / Laundry Area

- **Dimensions:** 1.5m × 2.5m (3.75 sqm)
- **Location:** Rear (connected to kitchen)
- **Features:**
  - Washing area
  - Good ventilation (semi-open)

Circulation and Layout Design

- **Main Entrance → Living Room → Dining → Kitchen**
- Bedrooms are placed **away from entrance** for privacy
- T&B is **centrally located** for equal access
- Service area is **separated** to avoid disturbance

COMPARISON AND ANALYSIS

The comparison between the original prompt and the improved prompt shows how important prompt engineering is in generating high-quality AI outputs. The original prompt, “Create a floor plan for a bungalow house,” was very simple and lacked specific instructions. Because of this, the AI generated a basic and generalized response. While it included standard parts of a house such as bedrooms, a kitchen, and a living area, it did not provide detailed information such as room dimensions, layout organization, or design considerations. This made the output less useful, especially for academic or engineering purposes.

On the other hand, the improved prompt was more detailed and specific. It included the total floor area (60–80 sqm), required number of rooms, and additional features like a service area. It also instructed the AI to provide dimensions, proper layout, and considerations for ventilation and efficiency. Because of these clear instructions, the AI produced a more structured and informative output. The result included labeled spaces, approximate measurements, and a logical arrangement of rooms based on function. It also considered real-world design principles such as zoning and accessibility.

This demonstrates that better prompts lead to better results. By adding constraints, details, and context, the AI is guided to produce outputs that are more accurate, relevant, and useful. In engineering applications, this is especially important because precision and clarity are required. The improved prompt transformed a simple answer into a more professional and practical floor plan concept. Overall, this activity shows that prompt engineering is a valuable skill when using AI tools. A well-constructed prompt allows users to maximize the potential of AI and obtain outputs that meet specific needs.